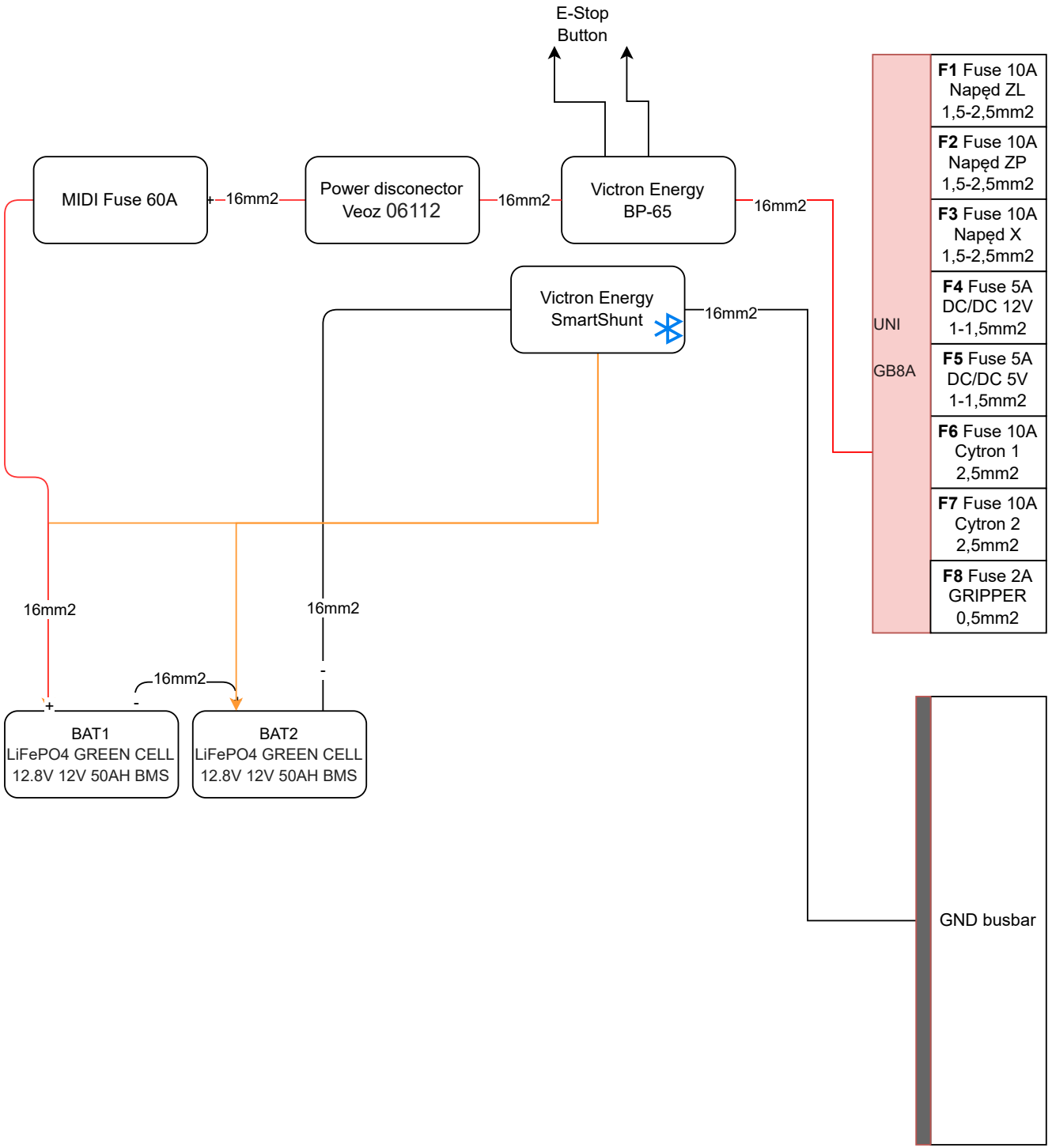
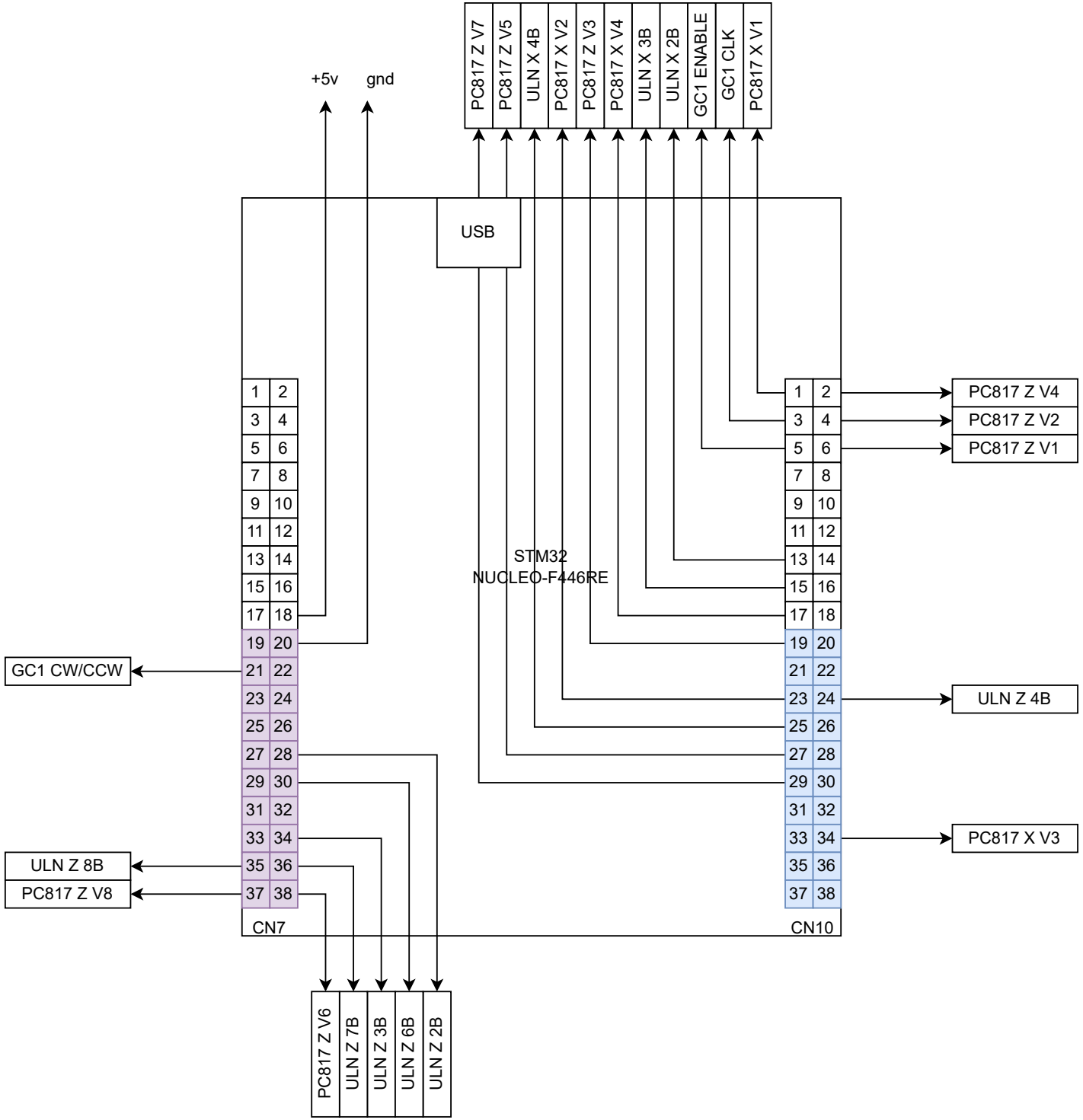
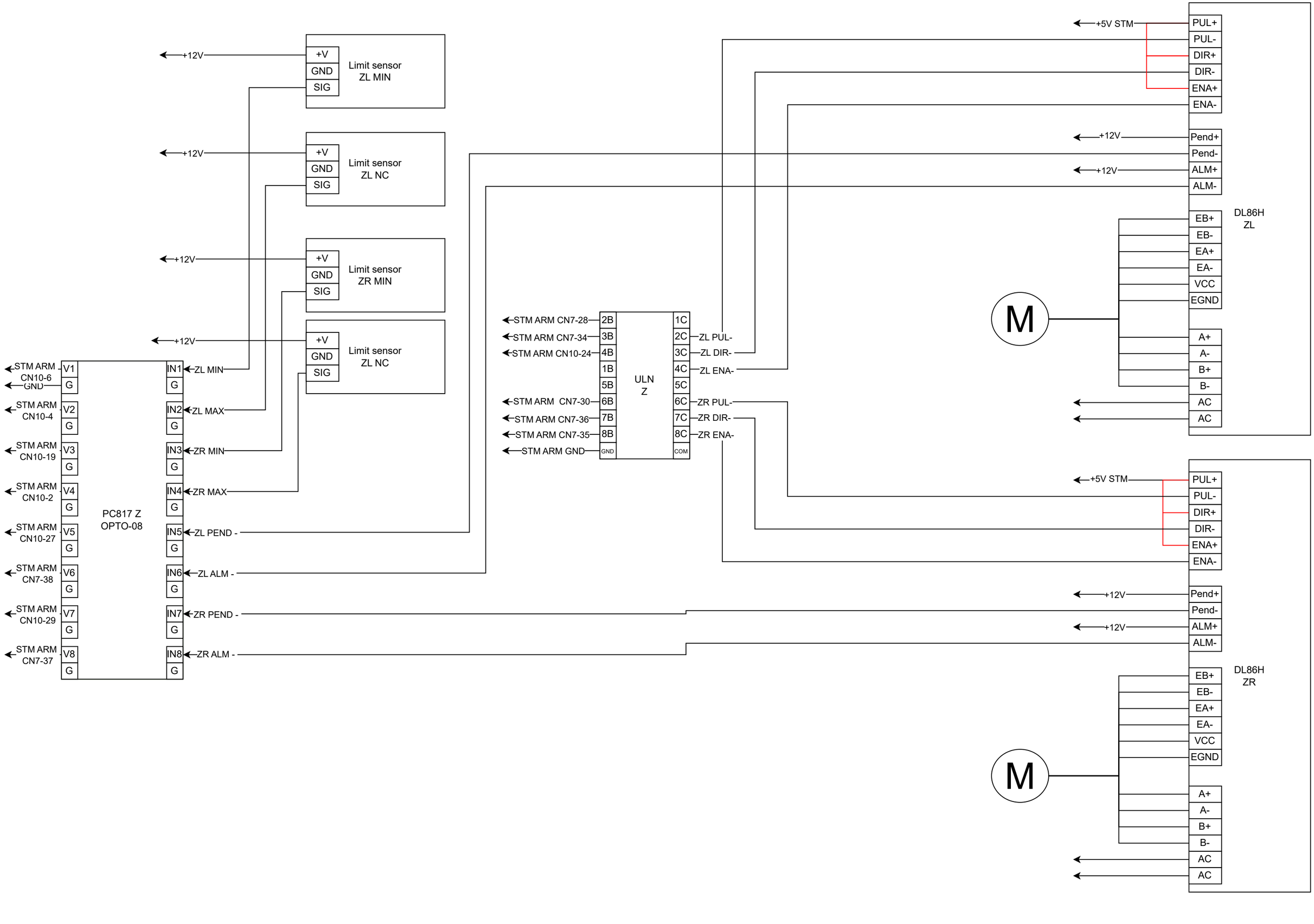
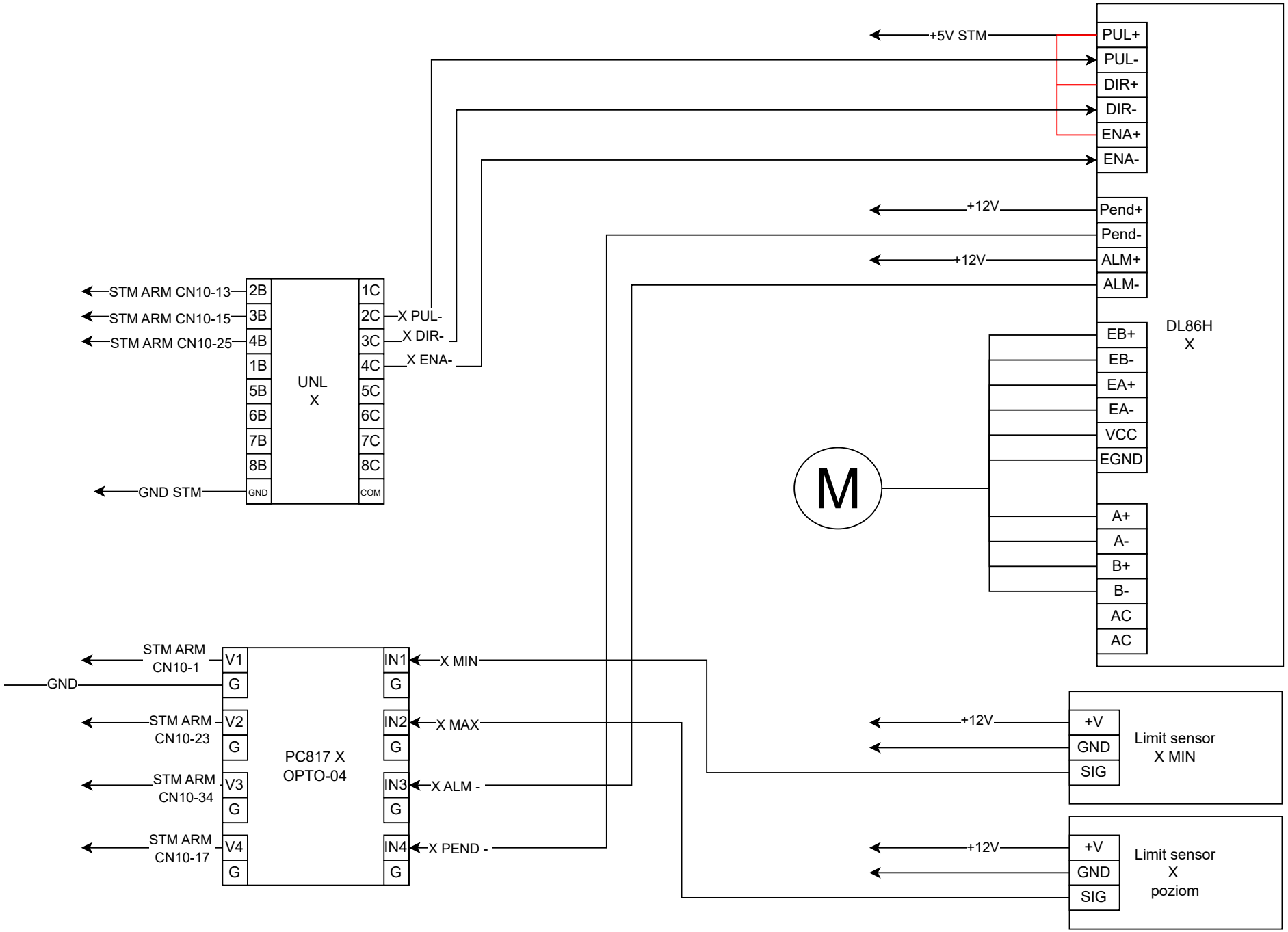
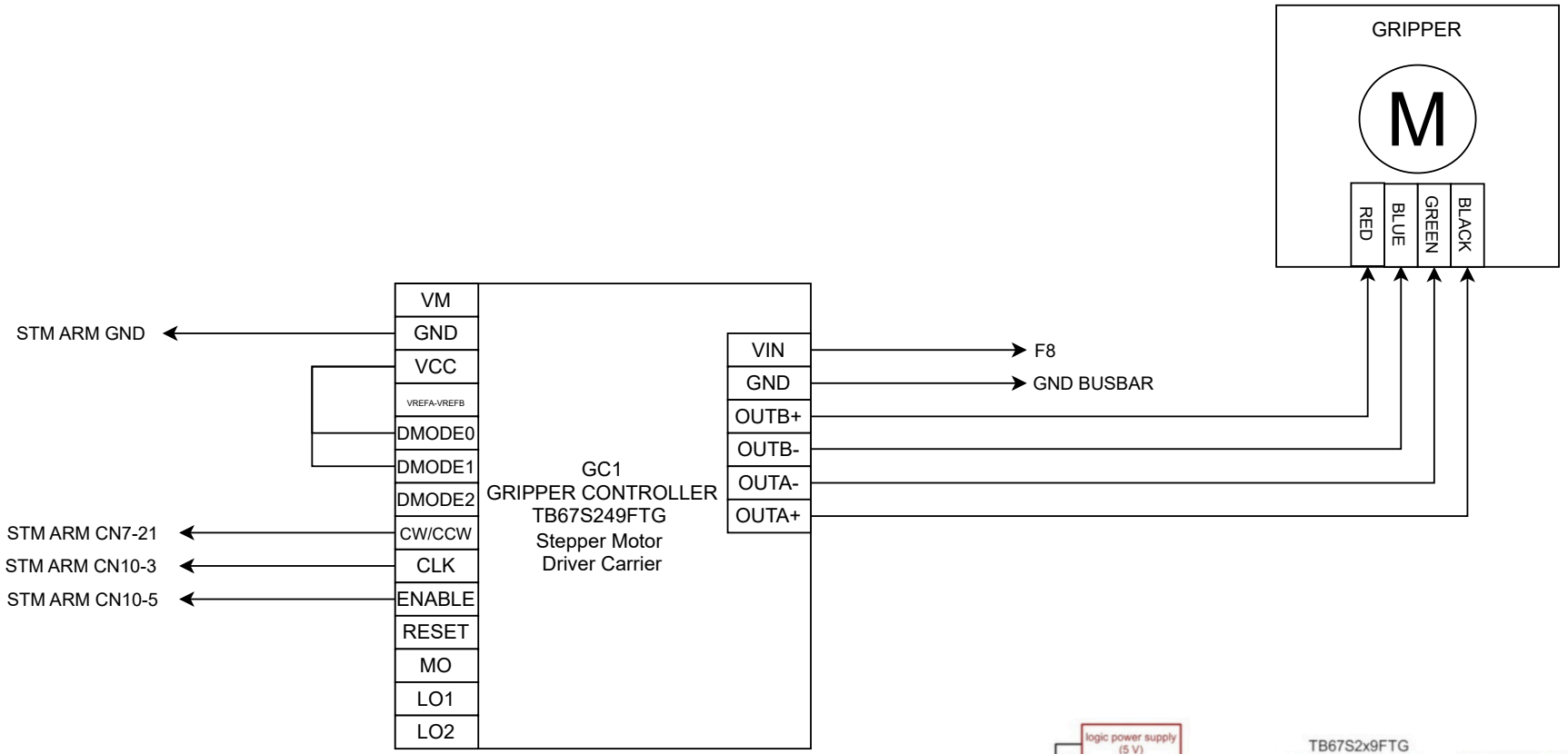
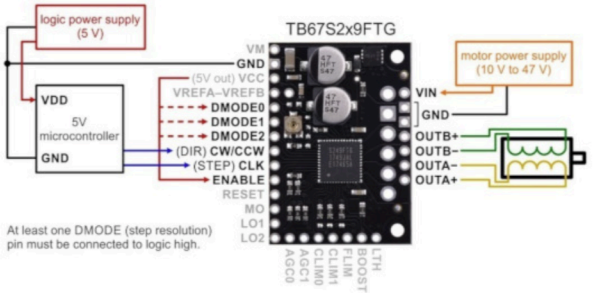

E-Stop
Button
MIDI Fuse 60A
16mm2
Power disconector
Veoz 06112
16mm2
Victron Energy
BP-65
16mm2
Victron Energy
SmartShunt
16mm2
UNI
GB8A
F1 Fuse 10A
Napęd ZL
1,5-2,5mm2
F2 Fuse 10A
Napęd ZP
1,5-2,5mm2
F3 Fuse 10A
Napęd X
1,5-2,5mm2
F4 Fuse 5A
DC/DC 12V
1-1,5mm2
F5 Fuse 5A
DC/DC 5V
1-1,5mm2
F6 Fuse 10A
Cytron 1
2,5mm2
F7 Fuse 10A
Cytron 2
2,5mm2
F8 Fuse 2A
GRIPPER
0,5mm2
16mm2
16mm2
16mm2
+
-
-
BAT1
LiFePO4 GREEN CELL
12.8V 12V 50AH BMS
BAT2
LiFePO4 GREEN CELL
12.8V 12V 50AH BMS
GND busbar

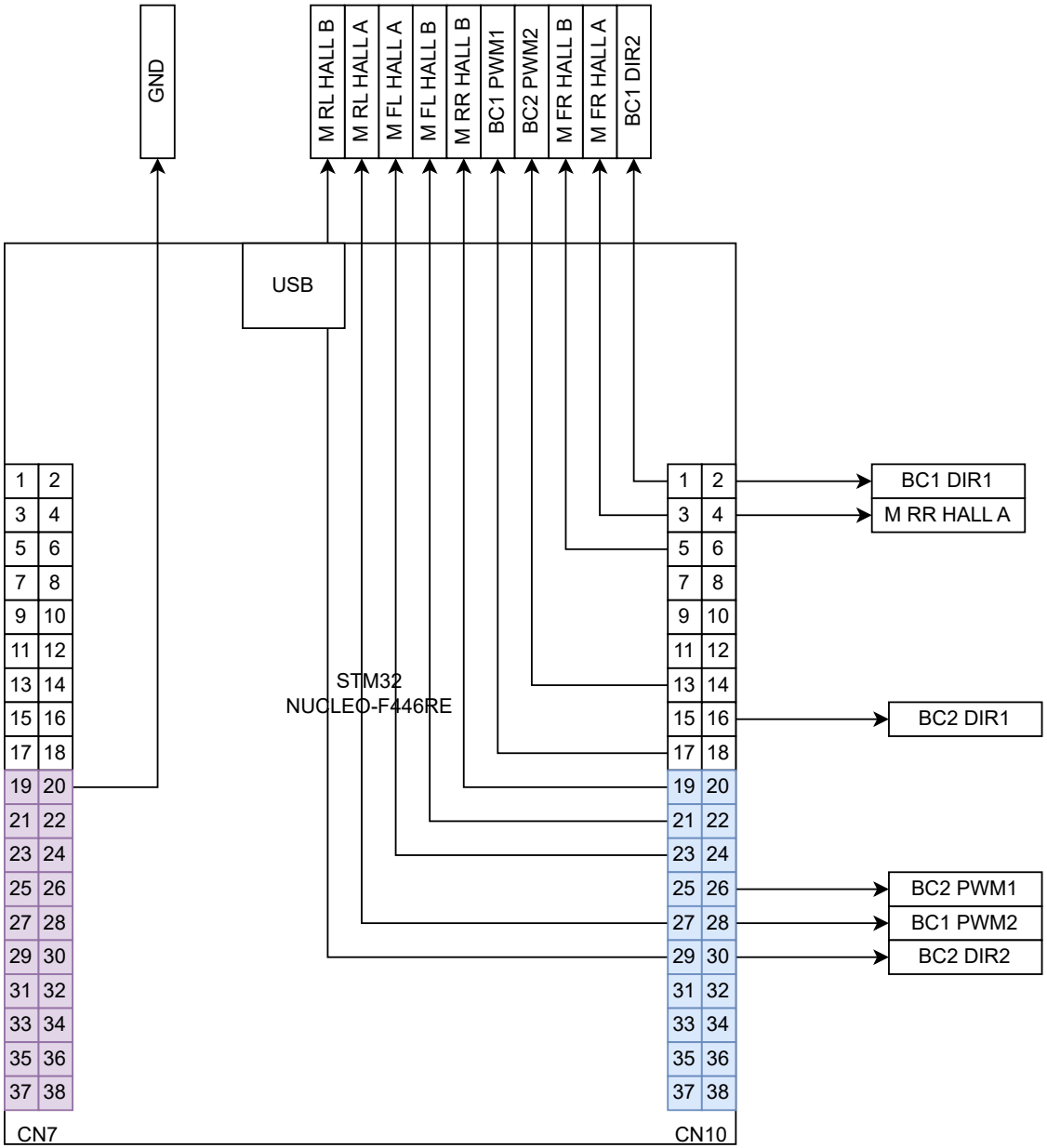

PC817 Z V7
PC817 Z V5
ULN X 4B
PC817 X V2
PC817 Z V3
PC817 X V4
ULN X 3B
ULN X 2B
GC1 ENABLE
GC1 CLK
PC817 X V1
+5v
gnd
USB
STM32
NUCLEO-F446RE
CN7
CN10
GC1 CW/CCW
ULN Z 8B
PC817 Z V8
PC817 Z V4
PC817 Z V2
PC817 Z V1
ULN Z 4B
PC817 X V3
PC817 Z V6
ULN Z 7B
ULN Z 3B
ULN Z 6B
ULN Z 2B

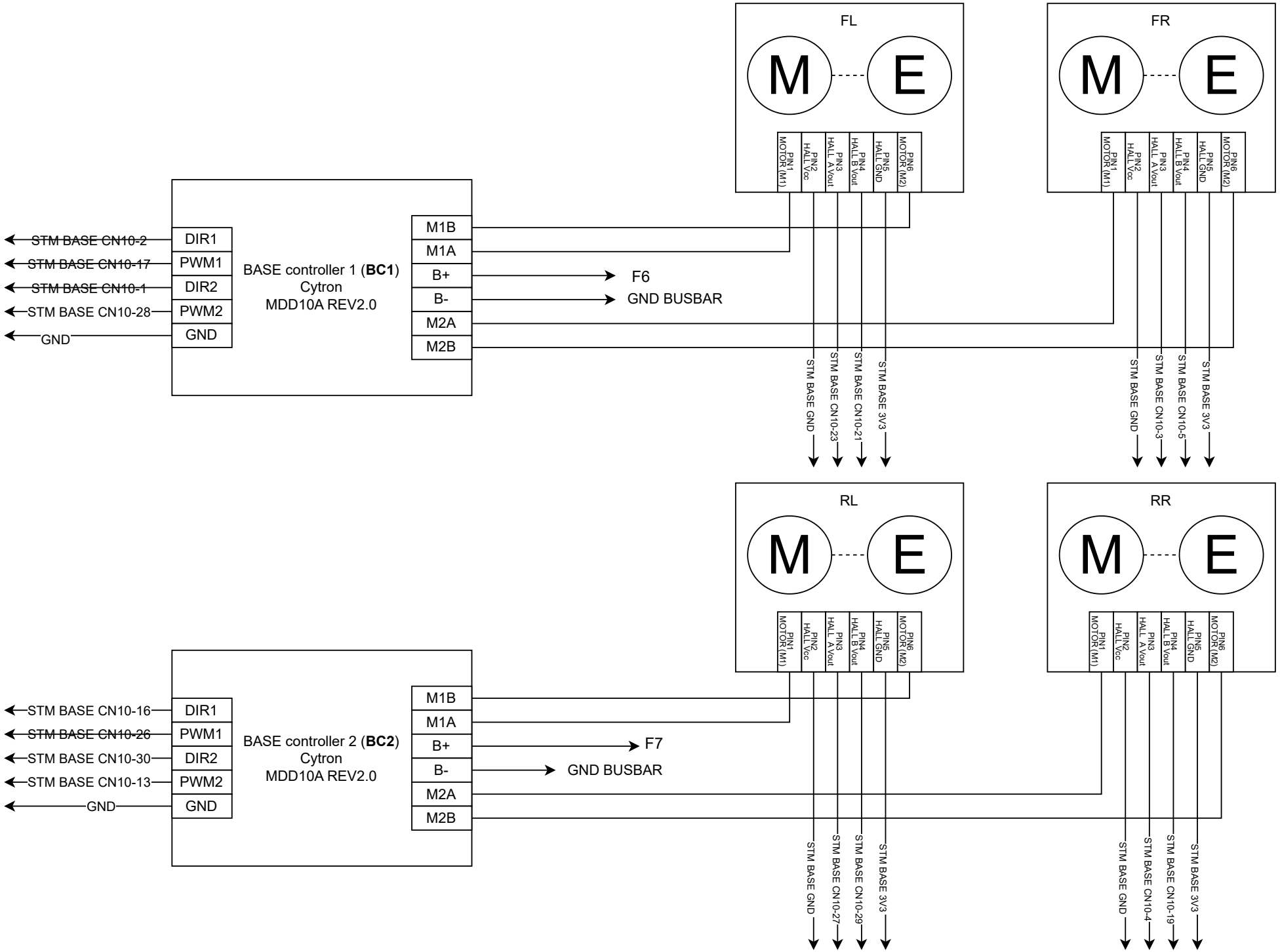

+12V
+V
GND
SIG
Limit sensor
ZL MIN
Limit sensor
ZL NC
Limit sensor
ZR MIN
Limit sensor
ZL NC
STM ARM
CN10-6
GND
STM ARM
CN10-4
STM ARM
CN10-19
STM ARM
CN10-2
STM ARM
CN10-27
STM ARM
CN7-38
STM ARM
CN10-29
STM ARM
CN7-37
V1
G
V2
V3
V4
V5
V6
V7
V8
IN1
IN2
IN3
IN4
IN5
IN6
IN7
IN8
ZL MIN
ZL MAX
ZR MIN
ZR MAX
ZL PEND -
ZL ALM -
ZR PEND -
ZR ALM -
PC817 Z
OPTO-08
STM ARM CN7-28
STM ARM CN7-34
STM ARM CN10-24
STM ARM CN7-30
STM ARM CN7-36
STM ARM CN7-35
STM ARM GND
2B
3B
4B
1B
5B
6B
7B
8B
GND
ULN
Z
1C
2C
3C
4C
5C
6C
7C
8C
COM
ZL PUL-
ZL DIR-
ZL ENA-
ZR PUL-
ZR DIR-
ZR ENA-
+5V STM
PUL+
PUL-
DIR+
DIR-
ENA+
ENA-
Pend+
Pend-
ALM+
ALM-
EB+
EB-
EA+
EA-
VCC
EGND
A+
A-
B+
B-
AC
DL86H
ZL
M
DL86H
ZR


+5V STM
PUL+
PUL-
DIR+
DIR-
ENA+
ENA-
+12V
Pend+
Pend-
+12V
ALM+
ALM-
DL86H
X
EB+
EB-
EA+
EA-
VCC
EGND
M
A+
A-
B+
B-
AC
AC
STM ARM CN10-13
STM ARM CN10-15
STM ARM CN10-25
2B
3B
4B
1B
5B
6B
7B
8B
GND
UNL
X
1C
2C
3C
4C
5C
6C
7C
8C
COM
X PUL-
X DIR-
X ENA-
GND STM
STM ARM CN10-1
GND
STM ARM CN10-23
STM ARM CN10-34
STM ARM CN10-17
V1
G
V2
G
V3
G
V4
G
PC817 X
OPTO-04
IN1
G
IN2
G
IN3
G
IN4
G
X MIN
X MAX
X ALM -
X PEND -
+12V
+V
GND
SIG
Limit sensor
X MIN
+12V
+V
GND
SIG
Limit sensor
X
poziom


GRIPPER
M
RED
BLUE
GREEN
BLACK
VM
GND
VCC
VREFA-VREFB
DMODE0
DMODE1
DMODE2
CW/CCW
CLK
ENABLE
RESET
MO
LO1
LO2
GC1
GRIPPER CONTROLLER
TB67S249FTG
Stepper Motor
Driver Carrier
VIN
GND
OUTB+
OUTB-
OUTA-
OUTA+
F8
GND BUSBAR
STM ARM GND
STM ARM CN7-21
STM ARM CN10-3
STM ARM CN10-5


TB67S2x9FTG
logic power supply (5 V)
motor power supply (10 V to 47 V)
VDD
5V microcontroller
GND
VM
GND
(5V out) VCC
VREFA–VREFB
DMODE0
DMODE1
DMODE2
(DIR) CW/CCW
(STEP) CLK
ENABLE
RESET
MO
LO1
LO2
VIN
GND
OUTB+
OUTB–
OUTA–
OUTA+
At least one DMODE (step resolution) pin must be connected to logic high.


GND
M RL HALL B
M RL HALL A
M FL HALL A
M FL HALL B
M RR HALL B
BC1 PWM1
BC2 PWM2
M FR HALL B
M FR HALL A
BC1 DIR2
USB
STM32
NUCLEO-F446RE
CN7
CN10
BC1 DIR1
M RR HALL A
BC2 DIR1
BC2 PWM1
BC1 PWM2
BC2 DIR2


FL
FR
RL
RR
M
E
PIN1 MOTOR (M1)
PIN2 HALL Vcc
PIN3 HALL A Vout
PIN4 HALL B Vout
PIN5 HALL GND
PIN6 MOTOR (M2)
BASE controller 1 (BC1)
Cytron
MDD10A REV2.0
DIR1
PWM1
DIR2
PWM2
GND
M1B
M1A
B+
B-
M2A
M2B
STM BASE CN10-2
STM BASE CN10-17
STM BASE CN10-1
STM BASE CN10-28
GND
F6
GND BUSBAR
STM BASE GND
STM BASE CN10-23
STM BASE CN10-21
STM BASE 3V3
STM BASE GND
STM BASE CN10-3
STM BASE CN10-5
STM BASE 3V3
BASE controller 2 (BC2)
Cytron
MDD10A REV2.0
STM BASE CN10-16
STM BASE CN10-26
STM BASE CN10-30
STM BASE CN10-13
GND
F7
GND BUSBAR
STM BASE GND
STM BASE CN10-27
STM BASE CN10-29
STM BASE 3V3
STM BASE GND
STM BASE CN10-4
STM BASE CN10-19
STM BASE 3V3